

Definitions in Analysis I by Terence Tao

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Remark. This document follows the third edition of *Analysis I*.

1 Introduction

No definitions found in this chapter.

2 Starting at the beginning: the natural numbers

2.1 The Peano axioms

Axiom 2.1. 0 is a natural number.

Axiom 2.2. If n is a natural number, then $n++$ is also a natural number.

Definition 2.1.3. We define 1 to be the number $0++$, 2 to be the number $(0++)++$, 3 to be the number $((0++)++)++$, etc. In other words, $1 := 0++$, $2 := 1++$, $3 := 2++$, etc.

Axiom 2.3. 0 is not the successor of any natural number, i.e., we have $n++ \neq 0$ for every natural number n .

Axiom 2.4. Different natural numbers must have different successors; i.e. if n, m are natural numbers and $n \neq m$, then $n++ \neq m++$. Equivalently, if $n++ = m++$, then we must have $n = m$.

Axiom 2.5 (Principle of mathematical induction). Let $P(n)$ be any property pertaining to a natural number n . Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n++)$ is also true. Then $P(n)$ is true for every natural number n .

2.2 Addition

Definition 2.2.1 (Addition of natural numbers). Let m be a natural number. To add 0 to m , we define $0+m := m$. Now, suppose inductively that we have defined how to add n to m . Then, we can add $n++$ to m by defining $(n++) + m := (n+m)++$.

Lemma 2.2.2. For any natural number n , $n + 0 = n$.

Lemma 2.2.3. For any natural numbers n and m , $n + (m++) = (n+m)++$.

Proposition 2.2.4 (Addition is commutative). For any natural numbers n and m , $n + m = m + n$.

Proposition 2.2.5 (Addition is associative). For any natural numbers a, b, c , we have $(a + b) + c = a + (b + c)$.

Proposition 2.2.6 (Cancellation law). Let a, b, c be natural numbers such that $a + b = a + c$. Then we have $b = c$.

Definition 2.2.7 (Positive natural numbers). A natural number b is said to be *positive* iff it is not equal to 0.

Proposition 2.2.8. If a is positive and b is a natural number, then $a + b$ is positive.

Corollary 2.2.9. If a and b are natural numbers such that $a + b = 0$, then $a = 0$ and $b = 0$.

Lemma 2.2.10. Let a be a positive number. Then there exists exactly one natural number b such that $b++ = a$.

Definition 2.2.11 (Ordering of the natural numbers). Let n and m be natural numbers. We say that n is greater than or equal to m , and write $n \geq m$ or $m \leq n$, iff we have $n = m + a$ for some natural number a . We say that n is *strictly greater than* m , and write $n > m$ or $m < n$, iff $n \geq m$ and $n \neq m$.

Proposition 2.2.12 (Basic properties of order for natural numbers). Let a, b, c be natural numbers. Then

- (a) *Reflexivity:* $a \geq a$.
- (b) *Transitivity:* If $a \geq b$ and $b \geq c$, then $a \geq c$.
- (c) *Anti-symmetry:* If $a \geq b$ and $b \geq a$, then $a = b$.
- (d) *Preservation of order:* $a \geq b$ if and only if $a + c \geq b + c$.
- (e) $a < b$ if and only if $a++ \leq b$.
- (f) $a < b$ if and only if $b = a + d$ for some positive number d .

Proposition 2.2.13 (Trichotomy of order for natural number). Let a and b be natural numbers. Then exactly one of the following statements is true: $a < b$, $a = b$, or $a > b$.

Proposition 2.2.14 (Strong principle of induction). Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m . Suppose that for each $m \geq m_0$, we have the following implication: if $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. Then we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.

2.3 Multiplication

Definition 2.3.1 (Multiplication of natural numbers). Let m be a natural number. To multiply 0 to m , we define $0 \times m := 0$. Now, suppose inductively that we have defined how to multiply n to m . Then, we can multiply $n++$ to m by defining $(n++) \times m := (n \times m) + m$.

Lemma 2.3.2 (Multiplication is commutative). Let n, m be natural numbers. Then $n \times m = m \times n$.

Lemma 2.3.3 (Positive natural numbers have no zero divisors). Let n, m be natural numbers. Then $n \times m = 0$ if and only if at least one of n, m is equal to 0. In particular, if n and m are both positive, then nm is also positive.

Proposition 2.3.4 (Distributive law). For any natural numbers a, b, c , we have $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

Proposition 2.3.5 (Multiplication is associative). For any natural numbers a, b, c , we have $(a \times b) \times c = a \times (b \times c)$.

Proposition 2.3.6 (Multiplication preserves order). If a, b are natural numbers such that $a < b$, and c is positive, then $ac < bc$.

Corollary 2.3.7 (Cancellation law). Let a, b, c be natural numbers such that $ac = bc$ and c is nonzero. Then $a = b$.

Proposition 2.3.9 (Euclidean algorithm). Let n be a natural number, and let q be a positive number. Then there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$.

Definition 2.3.11 (Exponentiation for natural numbers). Let m be a natural number. To raise m to the power 0, we define $m^0 := 1$; in particular, we define $0^0 := 1$. Now suppose recursively that m^n has been defined for some natural number n , then we define $m^{n++} := m^n \times m$.

3 Set theory

3.1 Fundamentals

Axiom 3.1 (Sets are objects). If A is a set, then A is also an object. In particular, given two sets A and B , it is meaningful to ask whether A is also an element of B .

Definition 3.1.4 (Equality of sets). Two sets A and B are *equal*, $A = B$, iff every element of A is an element of B and vice versa. To put it another way, $A = B$ if and only if every element x of A belongs also to B , and every element y of B belongs also to A .

Axiom 3.2 (Empty set). There exists a set $\emptyset$, known as the *empty set*, which contains no elements, i.e., for every object x we have $x \notin \emptyset$.

Lemma 3.1.6 (Single choice). Let A be a non-empty set. Then there exists an object x such that $x \in A$.

Axiom 3.3 (Singleton sets and pair sets). If a is an object, then there exists a set $\{a\}$ whose only element is a , i.e., for every object y , we have $y \in \{a\}$ if and only if $y = a$; we refer to $\{a\}$ as the singleton set whose element is a . Furthermore, if a and b are objects, then there exists a set $\{a, b\}$ whose only elements are a and b ; i.e., for every object y , we have $y \in \{a, b\}$ if and only if $y = a$ or $y = b$; we refer to this set as the pair set formed by a and b .

Axiom 3.4 (Pairwise union). Given any two sets A, B , there exists a set $A \cup B$, called the *union* $A \cup B$ of A and B , whose elements consist of all the elements which belong to A , or B or both. In other words, for any object x ,

$$x \in A \cup B \iff (x \in A \text{ or } x \in B).$$

Lemma 3.1.13. If a and b are objects, then $\{a, b\} = \{a\} \cup \{b\}$. If A, B, C are sets, then the union operation is commutative (i.e., $A \cup B = B \cup A$) and associative (i.e., $(A \cup B) \cup C = A \cup (B \cup C)$). Also, we have $A \cup A = A \cup \emptyset = \emptyset \cup A = A$.

Definition 3.1.15 (Subsets). Let A, B be sets. We say that A is a *subset* of B , denoted $A \subseteq B$, iff every element of A is also an element of B , i.e.

$$\text{For any object } x, \quad x \in A \implies x \in B.$$

We say that A is a *proper subset* of B , denoted $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Proposition 3.1.18 (Sets are partially ordered by set inclusion). Let A, B, C be sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. If $A \subseteq B$ and $B \subseteq A$, then $A = B$. Finally, if $A \subsetneq B$ and $B \subsetneq C$, then $A \subsetneq C$.

Axiom 3.5 (Axiom of specification/axiom of separation). Let A be a set, and for each $x \in A$, let $P(x)$ be a property pertaining to x (i.e., $P(x)$ is either a true statement or a false statement). Then there exists a set, called $\{x \in A : P(x) \text{ is true}\}$ (or simply $\{x \in A : P(x)\}$ for short), whose elements are precisely the elements x in A for which $P(x)$ is true. In other words, for any object y ,

$$y \in \{x \in A : P(x) \text{ is true}\} \iff (y \in A \text{ and } P(y) \text{ is true}).$$

Definition 3.1.23 (Intersections). The *intersection* $S_1 \cap S_2$ of two sets is defined to be the set

$$S_1 \cap S_2 := \{x \in S_1 : x \in S_2\}.$$

Definition 3.1.27 (Difference set). Given two sets A and B , we define the set $A - B$ or $A \setminus B$ to be the set A with any elements of B removed:

$$A \setminus B := \{x \in A : x \notin B\}.$$

Proposition 3.1.28 (Sets form a boolean algebra). Let A, B, C be sets, and let X be a set containing A, B, C as subsets.

- (a) (Minimal element) We have $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.
- (b) (Maximal element) We have $A \cup X = X$ and $A \cap X = A$.
- (c) (Identity) We have $A \cap A = A$ and $A \cup A = A$.
- (d) (Commutativity) We have $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
- (e) (Associativity) We have $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.
- (f) (Distributivity) We have $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
- (g) (Partition) We have $A \cup (X \setminus A) = X$ and $A \cap (X \setminus A) = \emptyset$.
- (h) (De Morgan Laws) We have $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ and $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

Axiom 3.6 (Replacement). Let A be a set. For any object $x \in A$ and any object y , suppose we have a statement $P(x, y)$ pertaining to x and y , such that for each $x \in A$ there is at most one y for which $P(x, y)$ is true. Then there exists a set $\{y : P(x, y) \text{ is true for some } x \in A\}$, such that for any object z ,

$$z \in \{y : P(x, y) \text{ is true for some } x \in A\} \iff P(x, z) \text{ is true for some } x \in A.$$

Axiom 3.7 (Infinity). There exists a set $\mathbb{N}$, whose elements are called natural numbers, as well as an object 0 in $\mathbb{N}$, and an object $n++$ assigned to every natural number $n \in \mathbb{N}$, such that the Peano axioms (Axioms 2.1 - 2.5) hold.

3.2 Russell's paradox

Axiom 3.8 (Universal specification). (*Dangerous!*) Suppose for every object x we have a property $P(x)$ pertaining to x (so that for every x , $P(x)$ is either a true statement or a false statement). Then there exists a set $\{x : P(x) \text{ is true}\}$ such that for every object y ,

$$y \in \{x : P(x) \text{ is true}\} \iff P(y) \text{ is true}.$$

Axiom 3.9 (Regularity). If A is a non-empty set, then there is at least one element x of A which is either not a set, or is disjoint from A .

3.3 Functions

Definition 3.3.1 (Functions). Let X, Y be sets, and let $P(x, y)$ be a property pertaining to an object $x \in X$ and an object $y \in Y$, such that for every $x \in X$, there is exactly one $y \in Y$ for which $P(x, y)$ is true (this is sometimes known as the *vertical line test*). Then we define the *function* $f : X \rightarrow Y$ *defined by* P *on the domain* X *and range* Y to be the object which, given any input $x \in X$, assigns an output $f(x) \in Y$, defined to be the unique object $f(x)$ for which $P(x, f(x))$ is true. Thus, for any $x \in X$ and $y \in Y$,

$$y = f(x) \iff P(x, y) \text{ is true}.$$

Definition 3.3.7 (Equality of functions). Two functions $f : X \rightarrow Y$, $g : X \rightarrow Y$ with the same domain and range are said to be equal, $f = g$, if and only if $f(x) = g(x)$

for all $x \in X$. (If $f(x)$ and $g(x)$ agree for some values of x , but not others, then we do not consider f and g to be equal.)

Definition 3.3.10 (Composition). Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two functions, such that the range of f is the same set as the domain of g . We then define the composition $g \circ f : X \rightarrow Z$ of the two functions g and f to be the function defined explicitly by the formula

$$(g \circ f)(x) := g(f(x)).$$

If the range of f does not match the domain of g , we leave the composition $g \circ f$ undefined.

Lemma 3.1.12 (Composition is associative). Let $f : Z \rightarrow W$, $g : Y \rightarrow Z$, and $h : X \rightarrow Y$ be functions. Then $f \circ (g \circ h) = (f \circ g) \circ h$.

Definition 3.3.14 (One-to-one function). A function f is *one-to-one* (or *injective*) if different elements map to different elements:

$$x \neq x' \implies f(x) \neq f(x').$$

Equivalently, a function is one-to-one if

$$f(x) = f(x') \implies x = x'.$$

Definition 3.3.17 (Onto functions). A function f is *onto* (or *surjective*) if $f(X) = Y$, i.e., every element in Y comes from applying f to some element in X :

For every $y \in Y$, there exists $x \in X$ such that $f(x) = y$.

Definition 3.3.20 (Bijective functions). Functions $f : X \rightarrow Y$ which are both one-to-one and onto are also called *bijective* or *invertible*.

3.4 Images and inverse images

Definition 3.4.1 (Images of sets). If $f : X \rightarrow Y$ is a function from X to Y , and S is a set in X , we define $f(S)$ to be the set

$$f(S) := \{f(x) : x \in S\};$$

this set is a subset of Y , and is sometimes called the *image* of S under the map f . In other words,

$$x \in S \implies f(x) \in f(S),$$

and

$$y \in f(S) \iff y = f(x) \text{ for some } x \in S.$$

Definition 3.4.4 (Inverse images). If U is a subset of Y , we define the set $f^{-1}(U)$ to be the set

$$f^{-1}(U) := \{x \in X : f(x) \in U\}.$$

In other words, $f^{-1}(U)$ consists of all the elements of X which map into U :

$$f(x) \in U \iff x \in f^{-1}(U).$$

We call $f^{-1}(U)$ the inverse image of U .

Axiom 3.10 (Power set axiom). Let X and Y be sets. Then there exists a set, denoted Y^X , which consists of all the functions from X to Y , thus

$$f \in Y^X \iff (f \text{ is a function with domain } X \text{ and range } Y).$$

Lemma 3.4.9. Let X be a set. Then the set

$$\{Y : Y \text{ is a subset of } X\}$$

is a set. This set is known as the *power set* of X and is denoted 2^X .

Axiom 3.11 (Union). Let A be a set, all of whose elements are themselves sets. Then there exists a set $\bigcup A$ whose elements are precisely those objects which are elements of the elements of A , thus for all objects x

$$x \in \bigcup A \iff (x \in S \text{ for some } S \in A).$$

If one has some set I , and for every element $\alpha \in I$ we have some set A_α , then we can form the union set $\bigcup_{\alpha \in I} A_\alpha$ by defining

$$\bigcup_{\alpha \in I} A_\alpha := \bigcup \{A_\alpha : \alpha \in I\}.$$

More generally, for any objects y ,

$$y \in \bigcup_{\alpha \in I} A_\alpha \iff (y \in A_\alpha \text{ for some } \alpha \in I). \quad (3.2)$$

We often refer to I as an *index set*, and the elements α of this index set as *labels*; the sets A_α are then called a *family of sets*, and are indexed by the labels $\alpha \in I$.

Unlabeled Definition (Intersection). Given any non-empty set I , and given an assignment of a set A_α to each $\alpha \in I$, we can define the intersection $\bigcap_{\alpha \in I} A_\alpha$ by first choosing some element β of I (which we can do since I is non-empty), and setting

$$\bigcap_{\alpha \in I} A_\alpha := \{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\}. \quad (3.3)$$

Then, for any object y ,

$$y \in \bigcap_{\alpha \in I} A_\alpha \iff (y \in A_\alpha \text{ for all } \alpha \in I). \quad (3.4)$$

3.5 Cartesian products

Definition 3.5.1 (Ordered pair). If x and y are any objects (possibly equal), we define the ordered pair (x, y) to be a new object, consisting of x as its first component and y as its second component. Two ordered pairs (x, y) and (x', y') are considered equal if and only if both their components match, i.e.

$$(x, y) = (x', y') \iff (x = x' \text{ and } y = y'). \quad (3.5)$$

Definition 3.5.4 (Cartesian product). If X and Y are sets, then we define the Cartesian product $X \times Y$ to be the collection of ordered pairs, whose first component lies in X and second component lies in Y , thus

$$X \times Y = \{(x, y) : x \in X, y \in Y\}$$

or equivalently

$$a \in (X \times Y) \iff (a = (x, y) \text{ for some } x \in X \text{ and } y \in Y).$$

Definition 3.5.7 (Ordered n -tuple and n -fold Cartesian product). Let n be a natural number. An ordered n -tuple $(x_i)_{1 \leq i \leq n}$ (also denoted $(x_1, \dots, x_n)$) is a collection of objects x_i , one for every natural number i between 1 and n ; we refer to x_i as the i^{th} component of the ordered n -tuple.

Two ordered n -tuples $(x_i)_{1 \leq i \leq n}$ and $(y_i)_{1 \leq i \leq n}$ are said to be equal iff $x_i = y_i$ for all $1 \leq i \leq n$.

If $(X_i)_{1 \leq i \leq n}$ is an ordered n -tuple of sets, we define their *Cartesian product* $\prod_{1 \leq i \leq n} X_i$ (also denoted $\prod_{i=1}^n X_i$ or $X_1 \times \dots \times X_n$) by

$$\prod_{1 \leq i \leq n} X_i := \{(x_i)_{1 \leq i \leq n} : x_i \in X_i \text{ for all } 1 \leq i \leq n\}.$$

If n is a natural number, we often write X^n as shorthand for the n -fold Cartesian product $X^n := \prod_{1 \leq i \leq n} X$.

Lemma 3.5.12 (Finite choice). Let $n \geq 1$ be a natural number, and for each natural number $1 \leq i \leq n$, let X_i be a non-empty set. Then there exists an n -tuple $(x_i)_{1 \leq i \leq n}$ such that $x_i \in X_i$ for all $1 \leq i \leq n$. In other words, if each X_i is non-empty, then the set $\prod_{1 \leq i \leq n} X_i$ is also non-empty.

3.6 Cardinality of sets

Definition 3.6.1 (Equal cardinality). We say that two sets X and Y have *equal cardinality* iff there exists a bijection $f : X \rightarrow Y$ from X to Y .

Proposition 3.6.4. Let X , Y , and Z be sets. Then X has equal cardinality with X . If X has equal cardinality with Y , then Y has equal cardinality with X . If X has equal cardinality with Y and Y has equal cardinality with Z , then X has equal cardinality with Z .

Definition 3.6.5. Let n be a natural number. A set X is said to have *cardinality* n , iff it has equal cardinality with $\{i \in \mathbb{N} : 1 \leq i \leq n\}$. We also say that X *has* n *elements* iff it has cardinality n .

Proposition 3.6.8 (Uniqueness of cardinality). Let X be a set with some cardinality n . Then X cannot have any other cardinality, i.e., X cannot have cardinality m for any $m \neq n$.

Lemma 3.6.9. Suppose that $n \geq 1$, and X has cardinality n . Then X is non-empty, and if x is any element of X , then the set $X - \{x\}$ (i.e., X with the element x removed) has cardinality $n - 1$.

Definition 3.6.10 (Finite sets). A set is *finite* iff it has cardinality N for some natural number n ; otherwise, the set is called *infinite*. If X is a finite set, we use $\#(X)$ to denote the cardinality of X .

Theorem 3.6.12. The set of natural numbers $\mathbb{N}$ is infinite.

Proposition 3.6.14 (Cardinal arithmetic).

- (a) Let X be a finite set, and let x be an object which is not an element of X . Then $X \cup \{x\}$ is finite and $\#(X \cup \{x\}) = \#(X) + 1$.
- (b) Let X and Y be finite sets. Then $X \cup Y$ is finite and $\#(X \cup Y) \leq \#(X) + \#(Y)$. If in addition X and Y are disjoint (i.e., $X \cap Y = \emptyset$), then $\#(X \cup Y) = \#(X) + \#(Y)$.
- (c) Let X be a finite set, and let Y be a subset of X . Then Y is finite, and $\#(Y) \leq \#(X)$. If in addition $Y \neq X$ (i.e., Y is a proper subset of X), then we have $\#(Y) < \#(X)$.
- (d) If X is a finite set, and $f : X \rightarrow Y$ is a function, then $f(X)$ is a finite set with $\#(f(X)) \leq \#(X)$. If in addition f is one-to-one, then $\#(f(X)) = \#(X)$.
- (e) Let X and Y be finite sets. Then Cartesian product $X \times Y$ is finite and $\#(X \times Y) = \#(X) \times \#(Y)$.
- (f) Let X and Y be finite sets. Then the set Y^X (defined in Axiom 3.10) is finite and $\#(Y^X) = \#(Y)^{\#(X)}$.

4 Integers and rationals

4.1 The integers

Definition 4.1.1 (Integers). An *integer* is an expression of the form $a - b$, where a and b are natural numbers. Two integers are considered to be equal, $a - b = c - d$, if and only if $a + d = c + b$. We let $\mathbb{Z}$ denote the set of all integers.

Definition 4.1.2. The sum of two integers, $(a - b) + (c - d)$, is defined by the formula

$$(a - b) + (c - d) := (a + c) - (b + d).$$

The product of two integers $(a - b) \times (c - d)$, is defined by

$$(a - b) \times (c - d) := (ac + bd) - (ad + bc).$$

Definition 4.1.4 (Negation of integers). If $(a - b)$ is an integer, we define the negation $-(a - b)$ to be the integer $(b - a)$. In particular, if $n = n - 0$ is a positive natural number, we can define its negation $-n = 0 - n$.

Lemma 4.1.5 (Trichotomy of integers). Let x be an integer. Then exactly one of the following three statements is true: (a) x is zero; (b) x is equal to a positive natural number n ; or (c) x is the negation $-n$ of a positive natural number n .

Unlabeled Definition. If n is a positive natural number, we call $-n$ a *negative integer*.

Proposition 4.1.6 (Laws of algebra for integers). Let x, y, z be integers. Then we have

$$\begin{aligned}x + y &= y + x \\(x + y) + z &= x + (y + z) \\x + 0 &= 0 + x = x \\x + (-x) &= (-x) + x = 0 \\xy &= yx \\(xy)z &= x(yz) \\x1 &= 1x = x \\x(y + z) &= xy + xz \\(y + z)x &= yx + zx.\end{aligned}$$

Remark. The above set of nine identities asserts that the integers form a *commutative ring*. If one deleted the identity $xy = yx$, then they would only assert that the integers form a *ring*.

Unlabeled Definition (Subtraction). Define the operation of *subtraction* $x - y$ of two integers by the formula

$$x - y := x + (-y).$$

Proposition 4.1.8 (Integers have no zero divisors). Let a and b be integers such that $ab = 0$. Then either $a = 0$ or $b = 0$ (or both).

Corollary 4.1.9 (Cancellation law for integers). If a, b, c are integers such that $ac = bc$ and c is non-zero, then $a = b$.

Definition 4.1.10 (Ordering of the integers). Let n and m be integers. We say that n is *greater than or equal to* m , and write $n \geq m$ or $m \leq n$, iff we have $n = m + a$ for some natural number a . We say that n is *strictly greater than* m , and write $n > m$ or $m < n$, iff $n \geq m$ and $n \neq m$.

Lemma 4.1.11 (Properties of order). Let a, b, c be integers.

- (a) $a > b$ if and only if $a - b$ is a positive natural number.
- (b) (Addition preserves order) If $a > b$, then $a + c > b + c$.
- (c) (Positive multiplication preserves order) If $a > b$ and c is positive, then $ac > bc$.
- (d) (Negation reverses order) If $a > b$, then $-a < -b$.
- (e) (Order is transitive) If $a > b$ and $b > c$, then $a > c$.
- (f) (Order trichotomy) Exactly one of the statements $a > b$, $a < b$, or $a = b$ is true.

4.2 The rationals

Definition 4.2.1. A *rational number* is an expression of the form a/b , where a and b are integers and b is non-zero; $a/0$ is not considered to be a rational number. Two

rational numbers are considered to be equal, $a//b = c//d$, if and only if $ad = cb$. The set of all rational numbers is denoted $\mathbb{Q}$.

Definition 4.2.2. If $a//b$ and $c//d$ are rational numbers, we define their sum

$$(a//b) + (c//d) := (ad + bc)//(bd),$$

their product

$$(a//b) \times (c//d) := (ac)//(bd),$$

and the negation

$$-(a//b) := (-a)//b.$$

Unlabeled Definition (Reciprocal). If $x = a//b$ is a non-zero rational (so that $a, b \neq 0$) then we define the *reciprocal* x^{-1} of x to be the rational number $x^{-1} := b//a$. We leave the reciprocal of 0 undefined.

Proposition 4.2.4 (Laws of algebra for rationals). Let x, y, z be rationals. Then the following laws of algebra hold:

$$\begin{aligned} x + y &= y + x \\ (x + y) + z &= x + (y + z) \\ x + 0 &= 0 + x = x \\ x + (-x) &= (-x) + x = 0 \\ xy &= yx \\ (xy)z &= x(yz) \\ x1 &= 1x = x \\ x(y + z) &= xy + xz \\ (y + z)x &= yx + zx. \end{aligned}$$

If x is non-zero, we also have

$$xx^{-1} = x^{-1}x = 1.$$

Remark. The above set of ten identities asserts that the rationals form a *field*.

Unlabeled Definition (Quotient). The *quotient* x/y of two rational numbers x and y , provided that y is non-zero, is defined by the formula

$$x/y = x \times y^{-1}.$$

Definition 4.2.6. A rational number x is said to be *positive* iff we have $x = a/b$ for some positive integers a and b . It is said to be *negative* iff we have $x = -y$ for some positive rational y (i.e. $x = (-a)/b$ for some positive integers a and b .)

Lemma 4.2.7 (Trichotomy of rationals). Let x be a rational number. Then exactly one of the following three statements is true: (a) x is equal to 0. (b) x is a positive rational number. (c) x is a negative rational number.

Definition 4.2.8 (Ordering of the rationals). Let x and y be rational numbers. We say that $x > y$ iff $x - y$ is a positive rational number, and $x < y$ iff $x - y$ is a negative rational number. We write $x \geq y$ iff either $x > y$ or $x = y$, and similarly define $x \leq y$.

Proposition 4.2.9 (Basic properties of order on the rationals).

- (a) (Order trichotomy) Exactly one of the three statements $x = y$, $x < y$, or $x > y$ is true.
- (b) (Order is anti-symmetric) One has $x < y$ if and only if $y > x$.
- (c) (Order is transitive) If $x < y$ and $y < z$, then $x < z$.
- (d) (Addition preserves order) If $x < y$, then $x + z < y + z$.
- (e) (Positive multiplication preserves order) If $x < y$ and z is positive, then $xz < yz$.

4.3 Absolute value and exponentiation

Definition 4.3.1 (Absolute value). If x is a rational number, the *absolute value* $|x|$ of x is defined as follows. If x is positive, then $|x| := x$. If x is negative, then $|x| := -x$. If x is zero, then $|x| := 0$.

Definition 4.3.2 (Distance). Let x and y be rational numbers. The quantity $|x - y|$ is called the *distance between x and y* and is sometimes denoted $d(x, y)$, thus $d(x, y) := |x - y|$. For instance, $d(3, 5) = 2$.

Proposition 4.3.3 (Basic properties of absolute value and distance). Let x, y, z be

rational numbers.

- (a) (Non-degeneracy of absolute value) We have $|x| \geq 0$. Also, $|x| = 0$ if and only if x is 0.
- (b) (Triangle inequality for absolute value) We have $|x + y| \leq |x| + |y|$.
- (c) We have the inequalities $-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, we have $-|x| \leq x \leq |x|$.
- (d) (Multiplicativity of absolute value) We have $|xy| = |x||y|$. In particular, $|-x| = |x|$.
- (e) (Non-degeneracy of distance) We have $d(x, y) \geq 0$. Also, $d(x, y) = 0$ if and only if $x = y$.
- (f) (Symmetry of distance) $d(x, y) = d(y, x)$.
- (g) (Triangle inequality for distance) $d(x, z) \leq d(x, y) + d(y, z)$.

Definition 4.3.4 (ε -closeness). Let $\varepsilon > 0$ be a rational number, and let x, y be rational numbers. We say that y is ε -close to x iff we have $d(y, x) \leq \varepsilon$.

Proposition 4.3.7. Let x, y, z, w be rational numbers.

- (a) If $x = y$, then x is ε -close to y for every $\varepsilon > 0$. Conversely, if x is ε -close to y for every $\varepsilon > 0$, then we have $x = y$.
- (b) Let $\varepsilon > 0$. If x is ε -close to y , then y is ε -close to x .
- (c) Let $\varepsilon, \delta > 0$. If x is ε -close to y , and y is δ -close to z , then x and z are $(\varepsilon + \delta)$ -close.
- (d) Let $\varepsilon, \delta > 0$. If x and y are ε -close, and z and w are δ -close, then $x + z$ and $y + w$ are $(\varepsilon + \delta)$ -close, and $x - z$ and $y - w$ are also $(\varepsilon + \delta)$ -close.
- (e) Let $\varepsilon > 0$. If x and y are ε -close, they are also ε' -close for every $\varepsilon' > \varepsilon$.

- (f) Let $\varepsilon > 0$. If y and z are both ε -close to x , and w is between y and z (i.e., $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε -close to x .
- (g) Let $\varepsilon > 0$. If x and y are ε -close, and z is non-zero, then xz and yz are $\varepsilon|z|$ -close.
- (h) Let $\varepsilon, \delta > 0$. If x and y are ε -close, and z and w are δ -close, then xz and yw are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$ -close.

Definition 4.3.9 (Exponentiation to a natural number). Let x be a rational number. To raise x to the power 0, we define $x^0 := 1$; in particular we define $0^0 := 1$. Now suppose inductively that x^n has been defined for some natural number n , then we define $x^{n+1} := x^n \times x$.

Proposition 4.3.10 (Properties of exponentiation, I). Let x, y be rational numbers, and let n, m be natural numbers.

- (a) We have $x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.
- (b) Suppose $n > 0$. Then we have $x^n = 0$ if and only if $x = 0$.
- (c) If $x \geq y \geq 0$, then $x^n \geq y^n \geq 0$. If $x > y \geq 0$ and $n > 0$, then $x^n > y^n \geq 0$.
- (d) We have $|x^n| = |x|^n$.

Definition 4.3.11 (Exponentiation to a negative number). Let x be a non-zero rational number. Then for any negative integer $-n$, we define $x^{-n} := 1/x^n$.

Proposition 4.3.12 (Properties of exponentiation, II). Let x, y be non-zero rational numbers, and let n, m be integers.

- (a) We have $x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.
- (b) If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive, and $0 < x^n \leq y^n$ if n is negative.
- (c) If $x, y > 0$, $n \neq 0$, and $x^n = y^n$, then $x = y$.
- (d) We have $|x^n| = |x|^n$.

4.4 Gaps in the rational numbers

Proposition 4.4.1 (Interspersing of integers by rationals). Let x be a rational number. Then there exists an integer n such that $n \leq x < n+1$. In fact, this integer is unique (i.e., for each x there is only one n for which $n \leq x \leq n+1$). In particular, there exists a natural number N such that $N > x$ (i.e., there is no such thing as a rational number which is larger than all the natural numbers).

Remark 4.4.2. The integer N for which $n \leq x < n+1$ is sometimes referred to as the *integer part* of x and is sometimes denoted $n = \lfloor x \rfloor$.

Proposition 4.4.3 (Interspersing of rationals by rationals). If x and y are two rationals such that $x < y$, then there exists a third rational z such that $x < z < y$.

5 The real numbers

5.1 Cauchy sequences

Definition 5.1.1 (Sequences). Let m be an integer. A *sequence* $(a_n)_{n=m}^{\infty}$ of rational numbers is any function from the set $\{n \in \mathbb{Z} : n \geq m\}$ to $\mathbb{Q}$, i.e., a mapping which assigns to each integer n greater or equal to m , a rational number a_n . More informally, a sequence $(a_n)_{n=m}^{\infty}$ of rational numbers is a collection of rationals $a_m, a_{m+1}, a_{m+2}, \dots$

Definition 5.1.3 (ε -steadiness). Let $\varepsilon > 0$. A sequence $(a_n)_{n=0}^{\infty}$ is said to be ε -*steady* iff each pair a_j, a_k of sequence elements is ε -close for every natural number j, k . In other words, the sequence $a_0, a_1, a_2, \dots$ is ε -steady iff $d(a_j, a_k) \leq \varepsilon$ for all j, k .

Definition 5.1.6 (Eventual ε -steadiness). Let $\varepsilon > 0$. A sequence $(a_n)_{n=0}^{\infty}$ is said to be *eventually* ε -steady iff the sequence $a_N, a_{N+1}, a_{N+2}, \dots$ is ε -steady for some natural number $N \geq 0$. In other words, the sequence $a_0, a_1, a_2, \dots$ is eventually ε -steady iff there exists an $N \geq 0$ such that $d(a_j, a_k) \leq \varepsilon$ for all $j, k \geq N$.

Definition 5.1.8 (Cauchy sequences). A sequence $(a_n)_{n=0}^{\infty}$ of rational numbers is said to be a *Cauchy sequence* iff for every rational number $\varepsilon > 0$, the sequence $(a_n)_{n=0}^{\infty}$ is eventually ε -steady. In other words, the sequence $a_0, a_1, a_2, \dots$ is a Cauchy sequence iff for every $\varepsilon > 0$, there exists an $N \geq 0$ such that $d(a_j, a_k) \leq \varepsilon$ for all $j, k \geq N$.

Definition 5.1.12 (Bounded sequences). Let $M \geq 0$ be rational. A finite sequence $a_1, a_2, \dots, a_n$ is *bounded by* M iff $|a_i| \leq M$ for all $1 \leq i \leq n$. An infinite sequence $(a_n)_{n=1}^{\infty}$ is *bounded by* M iff $|a_i| \leq M$ for all $i \geq 1$. A sequence is said to be *bounded* iff it is bounded by M for some rational $M \geq 0$.

Lemma 5.1.14 (Finite sequences are bounded). Every finite sequence $a_1, a_2, \dots, a_n$ is bounded.

Lemma 5.1.15 (Cauchy sequences are bounded). Every Cauchy sequence $(a_n)_{n=1}^{\infty}$ is bounded.

5.2 Equivalent Cauchy sequences

Definition 5.2.1 (ε -close sequences). Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be two sequences,

and let $\varepsilon > 0$. We say that the sequence $(a_n)_{n=0}^\infty$ is ε -close to $(b_n)_{n=0}^\infty$ if a_n is ε -close to b_n for each $n \in \mathbb{N}$. In other words, the sequence $a_0, a_1, a_2, \dots$ is ε -close to the sequence $b_0, b_1, b_2, \dots$ iff $|a_n - b_n| \leq \varepsilon$ for all $n = 0, 1, 2, \dots$

Definition 5.2.3 (Eventually ε -close sequences). Let $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ be two sequences, and let $\varepsilon > 0$. We say that the sequence $(a_n)_{n=0}^\infty$ is *eventually ε -close* to $(b_n)_{n=0}^\infty$ iff there exists an $N \geq 0$ such that the sequences $(a_n)_{n=N}^\infty$ and $(b_n)_{n=N}^\infty$ are ε -close. In other words, $a_0, a_1, a_2, \dots$ is eventually ε -close to $b_0, b_1, b_2, \dots$ iff there exists an $N \geq 0$ such that $|a_n - b_n| \leq \varepsilon$ for all $n \geq N$.

Definition 5.2.6 (Equivalent sequences). Two sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are *equivalent* iff for each rational $\varepsilon > 0$, the sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are eventually ε -close. In other words, $a_0, a_1, a_2, \dots$ and $b_0, b_1, b_2, \dots$ are equivalent iff for every rational $\varepsilon > 0$, there exists an $N \geq 0$ such that $|a_n - b_n| \leq \varepsilon$ for all $n \geq N$.